

## B.Tech. Degree III Semester Examination November 2018

### IT/CS 15-1303 DISCRETE COMPUTATIONAL STRUCTURES (2015 Scheme)

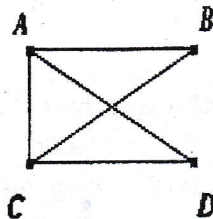
Time : 3 Hours

Maximum Marks : 60

#### PART A (Answer ALL questions)

(10 × 2 = 20)

- I. (a) Define converse, inverse and contrapositive of the conditional statement  $p \rightarrow q$ .
- (b) Prove that the relation *parallel* is an equivalence relation by defining it on the set of lines.
- (c) Let  $P = \{2, 3, 4, 5\}$  and R and S are two relations on P defined as  $R = \{(2,2), (2,3), (3,5), (4,2), (5,5)\}$ ,  $S = \{(2,3), (2,4), (3,4), (5,2)\}$ . Prove that  $R \circ S$  and  $S \circ R$  are not equal.
- (d) Define worst case, best case and average case time complexity of an algorithm.
- (e) How many 8 letter variable names can be formed from the alphabets 'a to i' with no letters repeated?
- (f) Draw a complete bi-partite graph  $K_{2,5}$ .
- (g) Give the memory representation for a multigraph using suitable example.
- (h) Draw the spanning trees of the given graph.



- (i) Define homomorphism and isomorphism of algebraic structure with example.
- (j) Let  $A = \{5, 6, 7, 8, 9, 10\}$  with partial ordering  $\leq$ . Draw the Hasse diagram and find the maximal and minimal element of A.

#### PART B

(4 × 10 = 40)

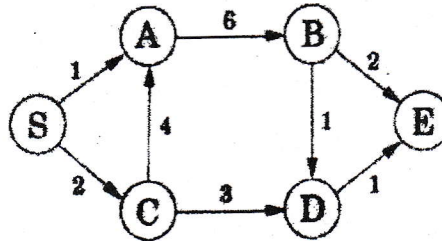
- II. (a) Prove that  $p \rightarrow q$ ,  $\bar{p} \vee q$ ,  $\bar{q} \rightarrow \bar{p}$  are equivalent. (5)
  - (b) Prove the validity of the given statements without using truth table "If 6 is even then 2 does not divide 7. Either 5 is not prime or 2 divides 7. But 5 is prime. Therefore 6 is even". (5)
- OR
- III. (a) Prove by mathematical induction  $n^3 + 2n$  is divisible by 3 for any integer  $n$ . (5)
  - (b) Define surjective function, injective function and bijective function with examples. (5)

(P.T.O.)

- IV. (a) What are the characteristics of algorithm? (5)  
 (b) Prove that if 9 colours are used to paint 100 houses, at least 12 houses will be of the same colour. (5)

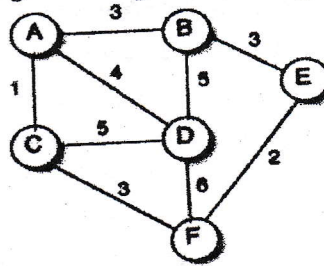
OR

- V. (a) Solve the recurrence equation  $T_r - 4T_{r-1} + 4T_{r-2} = 0$  with initial conditions  $T_0 = 1$  and  $T_1 = 6$ . (5)  
 (b) How many different 8 bit string are there that begin and end with 0? (5)
- VI. (a) Define regular graph. Draw a 2 regular graph with 3 vertices and 4 vertices. (4)  
 (b) Find the shortest path from node S to E using Dijkstra's algorithm. (6)



OR

- VII. (a) Define a complete binary tree and draw a complete binary tree. (4)  
 (b) Find the minimum spanning tree using Kruskal's algorithm for the given graph. (6)



- VIII. Define field. Check whether the set of odd integers on the operations addition and multiplication is a field. Also check for even integers. (10)

OR

- IX. (a) Consider the set  $D_{50} = \{1, 2, 4, 5, 10, 20, 25, 50\}$  and the partial ordering divisibility. Draw the Hasse diagram and find the supremum and infimum, upper bounds, lower bounds of the set  $B = \{5, 10\}$ . (8)  
 (b) Define ring with example. (2)

\*\*\*

